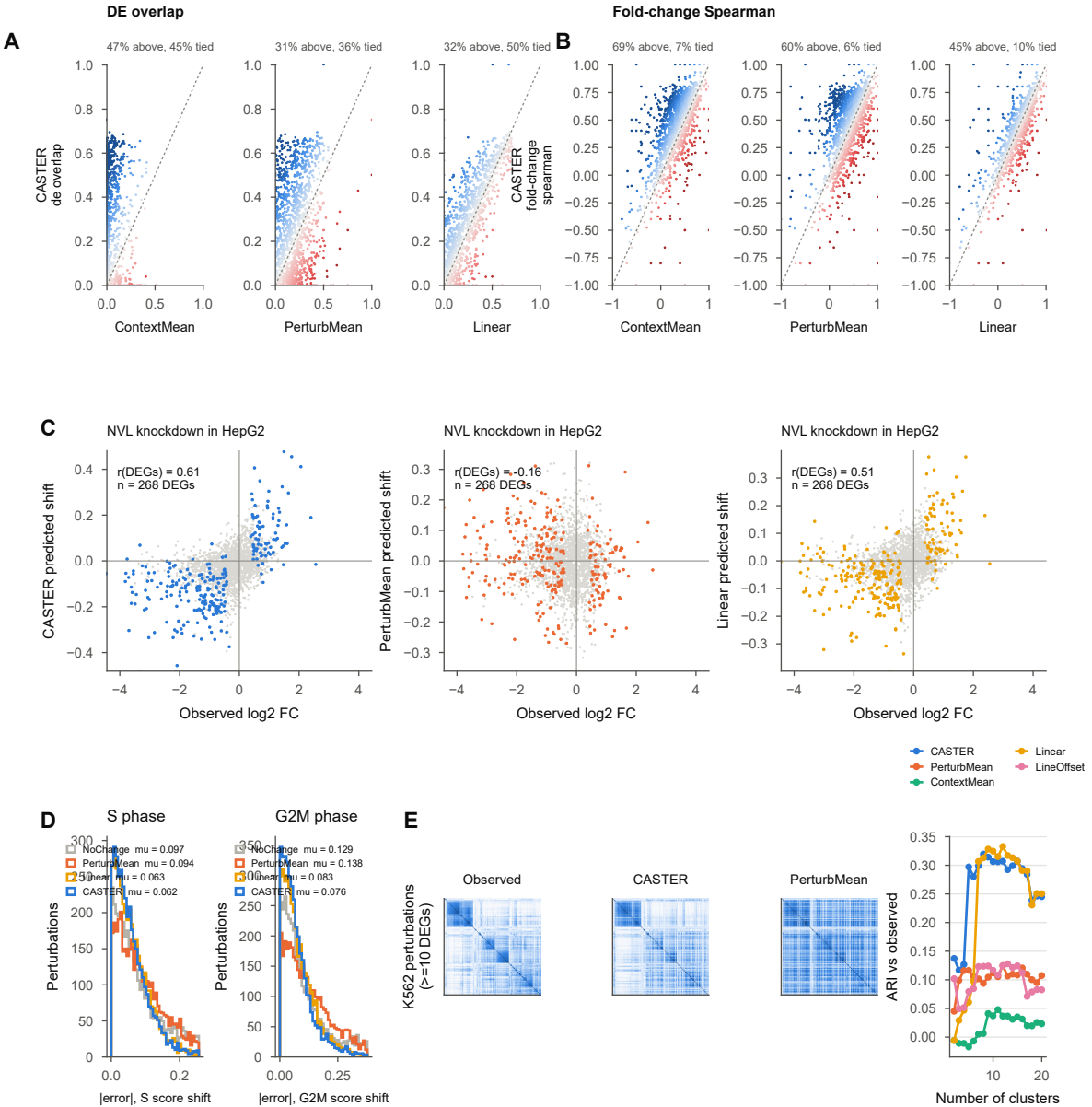

A
DE overlap
47% above, 45% tied
31% above, 36% tied
32% above, 50% tied
CASTER de overlap
ContextMean
PerturbMean
Linear
B
Fold-change Spearman
69% above, 7% tied
60% above, 6% tied
45% above, 10% tied
CASTER fold-change spearman
ContextMean
PerturbMean
Linear
C
NVL knockdown in HepG2
r(DEGs) = 0.61
n = 268 DEGs
CASTER predicted shift
Observed log2 FC
NVL knockdown in HepG2
r(DEGs) = -0.16
n = 268 DEGs
PerturbMean predicted shift
Observed log2 FC
NVL knockdown in HepG2
r(DEGs) = 0.51
n = 268 DEGs
Linear predicted shift
Observed log2 FC
CASTER
PerturbMean
ContextMean
Linear
LineOffset
D
S phase
NoChange mu = 0.097
PerturbMean mu = 0.094
Linear mu = 0.063
CASTER mu = 0.062
Perturbations
|error|, S score shift
G2M phase
NoChange mu = 0.129
PerturbMean mu = 0.138
Linear mu = 0.083
CASTER mu = 0.076
Perturbations
|error|, G2M score shift
E
Observed
CASTER
PerturbMean
K562 perturbations (>=10 DEGs)
ARI vs observed
Number of clusters